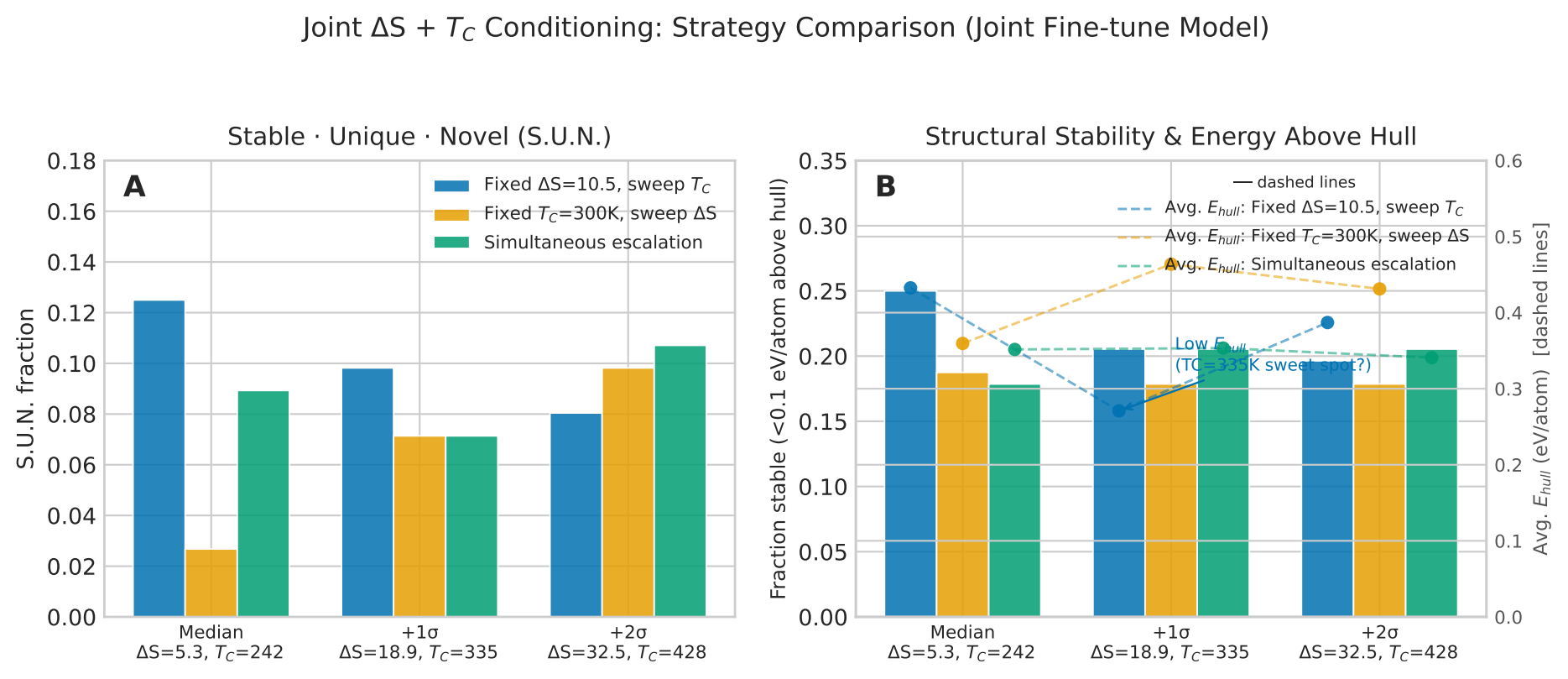
Joint ΔS + T_C Conditioning: Strategy Comparison (Joint Fine-tune Model)
Stable · Unique · Novel (S.U.N.)
A
Fixed ΔS=10.5, sweep T_C
Fixed T_C=300K, sweep ΔS
Simultaneous escalation
S.U.N. fraction
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Median
ΔS=5.3, T_C=242
+1σ
ΔS=18.9, T_C=335
+2σ
ΔS=32.5, T_C=428
Structural Stability & Energy Above Hull
B
dashed lines
Avg. E_{hull}: Fixed ΔS=10.5, sweep T_C
Avg. E_{hull}: Fixed T_C=300K, sweep ΔS
Avg. E_{hull}: Simultaneous escalation
Low E_{hull}
(TC=335K sweet spot?)
Fraction stable (<0.1 eV/atom above hull)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Avg. E_{hull} (eV/atom) [dashed lines]
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Median
ΔS=5.3, T_C=242
+1σ
ΔS=18.9, T_C=335
+2σ
ΔS=32.5, T_C=428